2024. 이수중3. 1중간

1 ④

$\sqrt{25}$의 제곱근 : $\pm\sqrt{5}$

2 ④

3 ④

$\sqrt{256}$의 제곱근 : ± 4

4 ⑤

수직선은 실수에 대응하는 점들로 완전히 메울 수 있다.

5 ② $\boxed{2-\sqrt{2}}$

$a=1,\quad b=\sqrt{2}-1$

$a-b=1-(\sqrt{2}-1)=2-\sqrt{2}$

6 ③

$\frac{6}{\sqrt{24}}=\frac{6}{2\sqrt{6}}=\frac{\sqrt{6}}{2}$

7 ② $\boxed{23.24}$

$\sqrt{540}=\sqrt{5.4\times100}=10\sqrt{5.4}=23.24$

8 ③

$\sqrt{\frac{75}{4}}=\frac{5\sqrt{3}}{2}$

9 ② $\boxed{-\frac{a}{8}}$

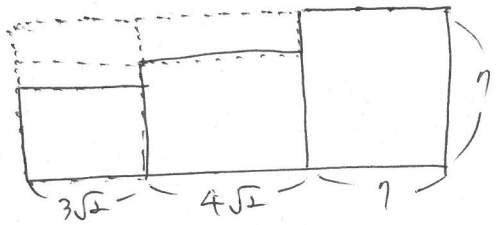
$\sqrt{\frac{a^2}{64}}=\frac{\sqrt{a^2}}{8}=\frac{-a}{8}=-\frac{a}{8}$

10 ⑤

11 ① $\boxed{2개}$

무리수 : $\sqrt{5}-2,\quad \sqrt{6.4}$

12 ② $\boxed{(28+14\sqrt{2})\,cm}$



둘레의 길이 $=2(3\sqrt{2}+4\sqrt{2}+7+7)$

$=28+14\sqrt{2}$

13 ③ $\boxed{\frac{3}{2}}$

$a^2=\frac{9}{4},\quad a=\frac{3}{2}$

14 ① $\boxed{20}$

$(5x-2)(y+3)=5xy+15x-2y-6$

$A=5,\ B=15,\ A+B=20$

15 ③ $\boxed{3}$

$(2x-1)(3x+A)=6x^2+(2A-3)x-A$

$=6x^2+13x-2$

$A=2,\quad B=1,\quad A+B=3$

$\boxed{16}$ ②

$(-4+x)(-4-x) = 16-x^2$

$\boxed{17}$ ⑤ $\boxed{x-y}$

$5x^2y-5xy^2 = 5xy(x-y)$

$\boxed{18}$ ②

$\left(2x+\frac{y}{2}\right)^2 = 4x^2+2xy+\frac{y^2}{4}$

$\boxed{19}$ ④ $\boxed{6}$

$12x^2-x-6=(3x+2)(4x-3)$

$3x \quad +2$

$4x \quad -3$

$a+b+c+d = 3+2+4+(-3) = 6$

$\boxed{20}$ ①

② $6ab-3a^2 = 3a(2b-a)$

③ $x^2-10x+25 = (x-5)^2$

④ $x^2-4y^2 = (x+2y)(x-2y)$

⑤ $x^2+x-6 = (x+3)(x-2)$

$\boxed{21}$ ④ $\boxed{7}$

$7.5^2 \times 0.14 - 2.5^2 \times 0.14$

$= 0.14 \times (7.5^2 - 2.5^2)$

$= 0.14 \times (7.5+2.5)(7.5-2.5)$

$= 0.14 \times 10 \times 5 = 7$

$\boxed{A1}$ $\boxed{1}$

$(2x-y)^2-(x+y)(x-y)$

$= 4x^2-4xy+y^2-(x^2-y^2)$

$= 3x^2-4xy+2y^2$

$a+b+c = 3+(-4)+2 = 1$

$\boxed{A2}$ $\boxed{2}$

$x-4<0, \quad x-2>0$

$\sqrt{x^2-8x+16}+\sqrt{x^2-4x+4}$

$= \sqrt{(x-4)^2}+\sqrt{(x-2)^2}$

$= -(x-4)+(x-2) = 2$

$\boxed{A3}$ $\boxed{-5}$

$x^2+ax+20 = (x-4)(x-5)$

$x \quad -4$

$x \quad -5$

$a = -9$

$3x^2-13x+b = (x-4)(3x-1)$

$x \quad -4$

$3x \quad -1$

$b = 4$

$a+b = -5$

$\boxed{A4}$ $\boxed{1}$

$\frac{2020 \times 2021 - 2021}{2020^2-1}$

$= \frac{2021 \times (2020-1)}{(2020+1)(2020-1)} = \frac{2021}{2021} = 1$